

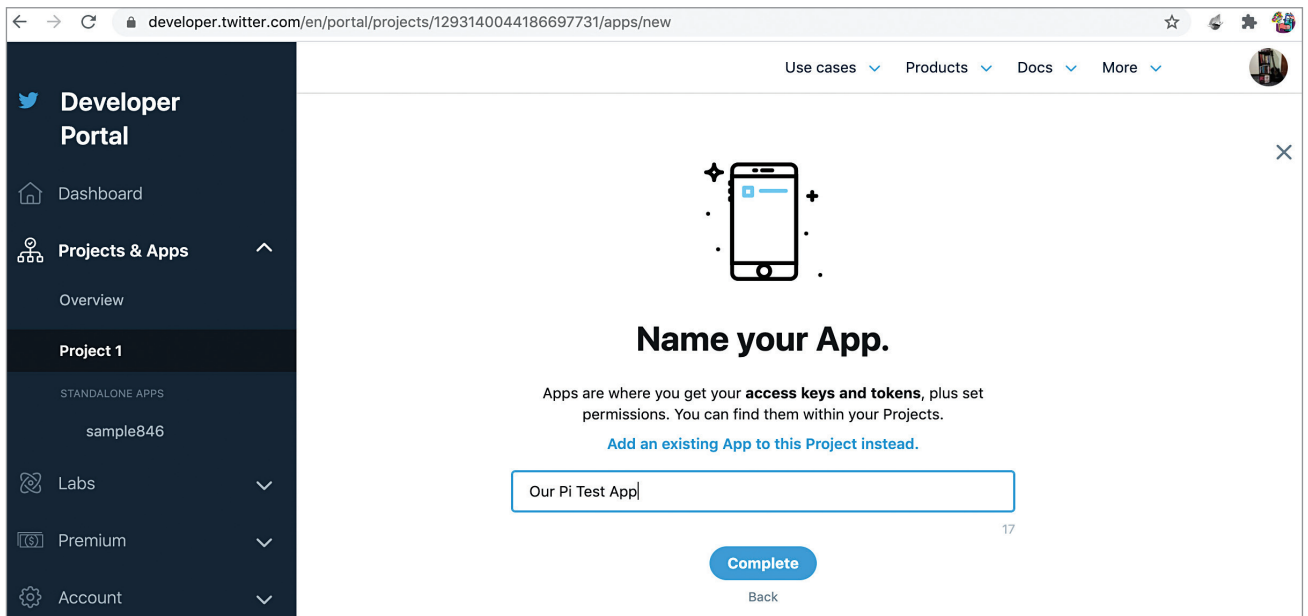


Fig. 1: Fill in the credentials on the Twitter New App Page

now monitor your various connected home appliances.

Raspberry Pi Bluetooth speakers

This project is a little bit more difficult if you do not have the right software tools; it is for this reason that we will use the Balena image (balenaSound) which will simplify our work. This can enable audio from Bluetooth and Spotify, as well as other synchronised audio.

What you need.

- Power supply and source
- MicroSD card
- Raspberry Pi
- Ethernet cable
- Bluetooth adaptor, if it is not already available

Setup steps. Download *balenaSound* image for use with Bluetooth. You have to have a Balena account to proceed.

(i) Create an application with the Balena Web interface.

(ii) On the screen that appears, choose the *Default Device Type* and select the Raspberry Pi that you will use as a default device. After that, click *Create and deploy*.

(iii) On the page you are taken to with the devices you have deployed, click on *Add device* to add Bluetooth speaker software, as it is not on the list.

(iv) A pop-up that will allow you to create a balenaOS image for the device you added in (iii) above appears. Select the device type; then choose the

version of the balenaOS image to use. For this purpose, we choose the *Production* option. Another option is the *Development* option, but this is used for troubleshooting purposes.

To use Wi-Fi, click the toggle under *Network Connection*; but you must also specify its SSID and passphrase.

(v) Now download balenaOS.

Adding the balenaOS to your Raspberry Pi. To add the balenaOS to your Raspberry Pi, we will use the Etcher tool to write the *balenaSound* image to the Raspberry Pi's SD card.

(i) In the Etcher software, click on the *Select Image* option. From the file dialogue that appears, select *balenaSound* image to write to the Raspberry SD card.

(ii) Choose the image you want to write the image to.

(iii) Click on *Flash* to write it to your SD card. Note that the drive you choose will be wiped.

After completion, you can now plug the drive into the Raspberry Pi to run.

Checking the status of your Raspberry Bluetooth speakers on Balena.

(i) Go to the Balena dashboard page and click the *balenaSound* application.

(ii) A list showing the devices running on this application will appear, and from it you can check the status of yours.

(iii) There is a *services* section from which you can see the Bluetooth audio service, Spotify, and Airplay.

From the interface, you can control the services by taking actions such as *start*, *restart* or *stop*.

Connecting your Raspberry Bluetooth speakers. There are three methods to play music using the speakers: Bluetooth, Airplay, and Spotify connect.

Through Bluetooth. Scan for the available Raspberry Pi Bluetooth speakers. Choose the one that appears as *balenaSound* and has a unique ID. With this, you can play music from your phone through the Raspberry Pi.

Through Airplay. BalenaSound has a built-in support capability for Apple AirPlay. This is used to send music from one device to another over a network. Find *balenaSound AirPlay* from the AirPlay devices list, given that it is followed by a unique ID.

Through Spotify Connect. Follow the same procedure as that for Bluetooth above by looking for *balenaSound* that has a unique identifier and choosing it.

Raspberry Pi Twitter Bot

This third project is a simple Twitter Bot built using Python as the programming language. The example below will show how to write the script while also implementing Twython.

What you need.

- MicroSD card
- Raspberry Pi 2/3
- Power supply
- Ethernet cord